

BY 527: Gene silencing and genome editing: principles and applications

Course Code	BY 527
Course Name	Gene silencing and genome editing: principles and applications
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. IDD 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	recommended by the 36th BoA (Sep 2020)

Course Contents

- **Introduction to gene structure and regulation**

Basic principles of genome organizations, gene structure, chromatin structure, gene silencing by histone modifications, gene silencing by DNA methylation (epigenetics and genome imprinting). Eukaryotic gene structure, eukaryotic transcription, transcriptional gene silencing. [6 Lectures]

- **RNA biology**

Post-transcriptional gene silencing. Introduction to RNAi (brief history and endogenous roles). Discovery of siRNAs and microRNAs. Molecular mechanisms RNAi. Genetic manipulations and RNAi in *C. elegans*, *Drosophila*, mammalian systems and plants. [6 Lectures]

- **Small silencing RNAs**

Classification, biogenesis and gene-regulatory mechanisms of small RNAs. Cellular functions of small RNAs. Functions of small RNAs in developmental biology, diseases and agriculture. RNAi screens (reverse genetic screens) in cell culture and model organisms. [6 Lectures]

- **Applications of RNAi**

RNAi-induced innate immunity and antiviral siRNAs. RNA-based therapeutics, vaccines and delivery of siRNAs. Examples of candidate RNAi drugs (Givosiran, Patisiran etc) [3 Lectures]

- **Genome editing methods-1**

Transgenesis, site-specific chromosomal integration by Cre-LoxP, phiC31-integrase, and Mos1-transposon. [6 Lectures]

- **Genome editing methods-2**

Genome engineering with TALENs and ZFNs. Discovery and mechanisms of CRISPR-Cas9 mediated genome editing. Different CRISPR systems and their uses in genome editing. [6 Lectures]

- **Module VII**

Designing of sgRNAs and repair templates. Next generation cloning technologies. Genome engineering methods for model organism. Construction of transgenics and knockouts using *C. elegans* model organism. [3 Lectures]

- **Applications of CRISPR mediated genome editing**

Cas9 for gene regulation: CRISPR interference (CRISPRi), CRISPR activation (CRISPRa) and CRISPRon. Genome-wide CRISPR knockout screens. Applications in agriculture, food and fuel industry. Ethical concerns of genome editing by CRISPR. [6 Lectures]

Text Books

1. Vijai Singh, Pawan K. Dhar, (Editors), **Genome Engineering via CRISPR-Cas9 System**, Academic Press, 2020.
2. Gregory J. Hannon (Editor), **RNAi: A Guide to Gene Silencing**, Cold Spring Harbor Laboratory Press, 2003.

References

1. Ute Schepers (Editor), **RNA Interference in Practice: Principles, Basics, and Methods for Gene Silencing in *C. elegans*, *Drosophila*, and Mammals**, Wiley-VCH Verlag GmbH, 2005.
2. Krishnarao Appasani, **MicroRNAs: From Basic Science to Disease Biology**, Cambridge University Press, 2008.
3. Rajesh K. Gaur, John J. Rossi, (Editors) **Regulation of Gene Expression by Small RNAs**, CRC Press, 2009.
4. Krishnarao Appasani (Editor), **Genome Editing and Engineering: From TALENs, ZFNs and CRISPRs to Molecular Surgery**, Cambridge University Press, 2008.
5. Jennifer Doudna (Editor), **CRISPR-Cas: A Laboratory Manual**, Cold Spring Harbor Laboratory Press, 2016.
6. Relevant research articles/reviews will be advised related to the topic being taught.